Exp : 14
7.10.26

Write a code using RAW sockets to implement packet sniffing.

```
from scapy.all import sniff
from scapy.layers.inet import IP, TCP, UDP, ICMP

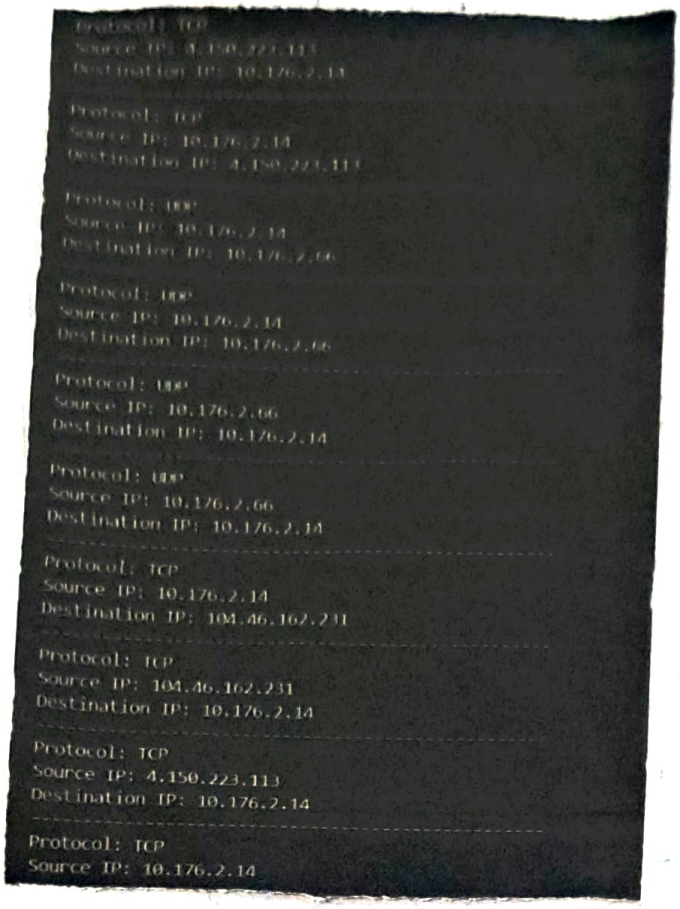
def packet_callback(packet):
    if IP in packet:
        ip_layer = packet[IP]
        protocol = ip_layer.proto
        src_ip = ip_layer.src
        dest_ip = ip_layer.dst

        # determine protocol
        protocol_name = ""
        if protocol == 1:
            protocol_name = "ICMP"
        elif protocol == 6:
            protocol_name = "TCP"
        elif protocol == 17:
            protocol_name = "UDP"
        else:
            protocol_name = "Unknown Protocol"

        # Print packet details
        print(f"Protocol: {protocol_name}")
        print(f"Source IP: {src_ip}")
        print(f"Destination IP: {dst_ip}")
        print("-" * 50)

# Capture packets on the default network interface
sniff(iface='wifi', prn=packet_callback,
      filter="ip", store=0)
```

O/P :



RESULT:

Thus a code using Raw sockets to implement packet sniffing was implemented.